

# Brandon Kochnari

## Software Engineer

Toronto, ON

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[BrandonKochnari](#)

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### EDUCATION

University of Toronto | Honours Bachelor of Science | Computer Science SPECIALIST Student

Expected Graduation: April 2028

**Relevant Courses:** Software Design, Data Structures & Analysis, Software Tools & Systems Programming, Foundations of Computer Science I & II, Introduction to the Theory of Computation

### TECHNICAL SKILLS

Languages: **Java** **Python** **C#** **C** **JavaScript** **TypeScript** **SQL**

Frontend: **React** **HTML** **Tailwind CSS** **CSS** **Vite**

Backend & Databases: **FastAPI** **PostgreSQL** **SQLAlchemy** **Alembic** **REST APIs** **JWT**

Tools: **Git** **GitHub** **Docker** **Vercel** **Render** **Linux** **CI/CD** **JetBrains IDEs** **Visual Studio**

**Core Concepts:** Full-Stack Development, Object-Oriented Programming (OOP), Data Structures & Algorithms (DS&A), Algorithm Design & Complexity, Software Engineering Principles, Testing & Debugging, Clean Architecture, SOLID Principles, DevOps

### PROJECTS

**URL Shortlink** — [github.com/BrandonKochnari/url-shortlink](#) — [www.urlshortlink.xyz](#) **May 2026 - Present**  
Software Developer

- **Key Focus:** Full-Stack, REST APIs, Authentication, Database Design, CI/CD
- Built a **full-stack URL shortener** in **Python** with **React**, **TypeScript**, **FastAPI**, and **PostgreSQL**, supporting **multiple user flows** including authenticated login, persistent links, click analytics, and redirect handling.
- Implemented **JWT authentication** and **protected REST API endpoints** to secure user accounts, enforce user-specific access, and manage link creation, expirations, activation status, redirects, and analytics.
- Designed **4+ relational database models** with **SQLAlchemy**, **PostgreSQL**, and **Alembic** to persist users, links, expirations, and analytics records.
- Deployed the frontend on **Vercel** and backend on **Render**, using **Docker** support, and **pytest API testing**.

**ChessPlus Computer Game** — [github.com/BrandonKochnari/ChessPlus](#) **Jul 2022 - Present**  
Lead Developer

- **Key Focus:** Game Design, Algorithmic Logic, OOP Design, Long-Term Development, Team Management
- Built a **Python/Pygame chess variant** in **object-oriented 8x8 board engine** with RPG-style HP, attacks, abilities, timers, and turn-based logic.
- Developed a **sprite-based UI** with menus, mouse selection, stat displays, and real-time chess clocks.
- Prototyped a **Unity/C# version** with **MonoBehaviour scripts**, prefab boards, **Unity UI**, and **input handling**.
- Scaled among **dozens of high school peers** by organizing play sessions, collecting feedback, and **iterating** on gameplay features to **improve replayability**.

**Finance Tracker** — [github.com/BrandonKochnari/finance-tracker](#) **Oct 2025 - Dec 2025**  
Software Developer

- **Key Focus:** Code Architecture, AI Integration, Use-Case Driven, Collaborative Development
- Built a **Java Swing finance tracker** with **Clean Architecture** and **SOLID Principles**, supporting budgets, transactions, labels, filtering, and spending analytics.
- Implemented monthly budgeting with balance calculations, **JSON persistence**, **layered controllers / presenters**, and **JUnit/Mockito tests**.
- Integrated the **Gemini API** to generate spending optimization advice from **user transaction data**.

### AWARDS

#### Ted Mossman Scholarship

Awarded \$1200 by the University of Toronto for demonstrating exceptional academic promise and performance.

### LANGUAGES (FLUENT)

English, French, Farsi

### EXTRACURRICULAR ACTIVITIES

#### High School Film Club

Co-Founder & President